LOCAL NOETHERIANNESS IS UNCHANGED BY BASE CHANGE ALONG A FINITELY GENERATED FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma records that the Noetherian property is preserved by base change along finitely generated field extensions and that local finite type over the base also suffices; those remarks lie outside the counted span.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-locally-Noetherian-base-change is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Local Noetherianness is local on affine open coverings. Properties, Lemma properties-lemma-locally-Noetherian, invoked at proof lines 2324-2325. Supplies the reduction to the affine case.
- Descent of Noetherianness along a faithfully flat ring map. Algebra, Lemma algebra-lemma-descent-Noetherian, invoked at proof lines 2328-2330. Supplies the converse direction.
- Base change of a Noetherian algebra along a finitely generated field extension stays Noetherian. Algebra, Lemma algebra-lemma-Noetherian-field-extension, invoked at proof lines 2331-2332. Supplies the forward direction.

Source reference. Original lemma 045I: sources/stacks-chow/tags.txt, source lines 5437–5437. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 033E: sources/stacks-chow/tags.txt, source lines 4053–4053. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OW: sources/stacks-chow/tags.txt, source lines 2209–2209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let k'/k be a finitely generated field extension. Then X is locally Noetherian if and only if $X_{k'}$ is locally Noetherian.*

Proof. Using Properties, Lemma 01OW we reduce to the case where X is affine, say $X = \mathrm{Spec}(A)$. In this case we have to prove that A is Noetherian if and only if $A_{k'}$ is Noetherian. Since $A \to A_{k'} = k' \otimes_k A$ is faithfully flat, we see that if $A_{k'}$ is Noetherian, then so is A, by Algebra, Lemma 033E. Conversely, if A is Noetherian then $A_{k'}$ is Noetherian by Algebra, Lemma 045I. □